IT Helpdesk Performance Analysis

A Power BI Project Report on SLA Compliance & Agent Performance

A self-initiated analytics project modeled on my experience as an Associate, IT Helpdesk — analyzing 11,923 support tickets across 10 countries to report on SLA compliance, ticket-handling efficiency, and agent performance.

Prepared by	Sreekar Tirunagari
Role focus	Business Analyst (background: Associate, IT Helpdesk)
Dataset period	January 2024 - May 2025
Tickets analyzed	11,923
Tools	Power BI Desktop, Power Query, DAX

Contents

1 Executive Summary
2 Data Model & Methodology
3 Key Performance Indicators
4 Ticket Volume Trend
5 Priority Analysis
6 Ticket Type Analysis
7 Queue & Workload Analysis
8 Geographic Distribution
9 Agent Performance
10 Customer Satisfaction
11 Key Findings & Recommendations
12 Appendix - Data Dictionary

1 Executive Summary

This project analyzes 11,923 IT helpdesk tickets logged between January 2024 and May 2025 across 10 countries, 10 support queues, and 12 agents. It was built as a self-directed Business Analyst portfolio piece, using the same reporting questions I supported daily as an Associate on an IT Helpdesk team: are we hitting our SLA commitments, where are we missing them, and how is each agent performing on speed and customer satisfaction.

11,923	75.7%	98.3%	56.0%	3.97 / 5
Total Tickets	Overall SLA Compliance	High-Priority SLA	Medium-Priority SLA	Avg. Customer Rating

Headline finding: overall SLA performance (75.7%) sits close to the 75% target, but this masks a sharp split by priority. High-priority tickets are resolved fast and reliably (98.3% SLA), while Medium-priority tickets — the largest volume segment at 42% of all tickets — fall well short at only 56.0% SLA compliance. Closing this gap is the single highest-leverage opportunity found in this analysis.

The sections that follow walk through the data model, then work through the dashboard one analytical lens at a time — volume trend, priority, ticket type, queue, geography, agents, and customer satisfaction — closing with findings and recommendations a support-operations manager could act on directly.

2 Data Model & Methodology

2.1 Data Model

The report is built on a star-schema semantic model in Power BI, with one fact table and several supporting dimension and helper tables:

Table	Role	Details
IT Helpdesk	Fact table	11,923 tickets; 29 columns incl. Priority, Type, Queue, Agent, Rating, Country
Calendar	Date dimension	One row per day, Jan 2024 - Dec 2025, set as the model's official date table
Country	Geo dimension	10 countries with flag/URL lookups for map visuals
Dictionary	Documentation	Column-level data dictionary used for in-report tooltips
KPI / Pages	Report control	Helper tables driving dynamic card titles and bookmark navigation

2.2 Derived Business Logic (Calculated Columns)

- **Resolution Hours** — whole hours between ticket creation and resolution.
- **Priority SLA** — target turnaround by priority: High 48h, Medium 72h, Low 96h.
- **SLA Status** — flags whether Resolution Hours met the Priority SLA target.
- **Severity** — derived urgency label combining Priority and Type (e.g. High-priority Incidents = Critical).

2.3 DAX Measures

42 DAX measures were built across six display folders to keep the measure list organized and easy to audit:

Folder	Examples
Core Metrics	Total Tickets, Avg Rating, Avg Resolution Days, % of Total Tickets
Priority Analysis	High/Medium/Low Priority Tickets, % High Priority
Type Analysis	Incident/Problem Tickets, Violations
SLA Analysis	Tickets Within/Outside SLA, % Within SLA, SLA Achievement Summary
Time Intelligence	Tickets MTD, Tickets Prior Month, MoM % Change, YoY % Change
Queue & Agent Analysis	Total Queues/Agents, Avg Tickets per Agent, Queue Rank by Volume

2.4 Workflow

Raw CSV export → Power Query cleaning (trimmed column names, type corrections, null handling on Rating) → star-schema model in Power BI → calculated columns for SLA/Severity logic → 42 DAX measures → 2-page interactive report with bookmark-driven navigation and dynamic titles.

3 Key Performance Indicators

A snapshot of the headline numbers behind the dashboard, pulled directly from the Power BI model:

11,923 Total Tickets	**10** Countries	**10** Queues	**12** Agents	**3.03 days** Avg. Resolution Time
75.7% Overall SLA	**38.3%** High-Priority Share	**2,022** Critical Tickets	**2,900** SLA Violations	**3.97 / 5** Avg. Rating

SLA targets applied in the model: **High priority = 48 hours, Medium = 72 hours, Low = 96 hours**, each measured against a 75% compliance goal.

4 Ticket Volume Trend

Monthly ticket volume has stayed in a fairly narrow band (617-740 tickets/month) across the full 17-month period, with no strong seasonal spike — this is a steady-state support operation rather than one with sharp demand swings.

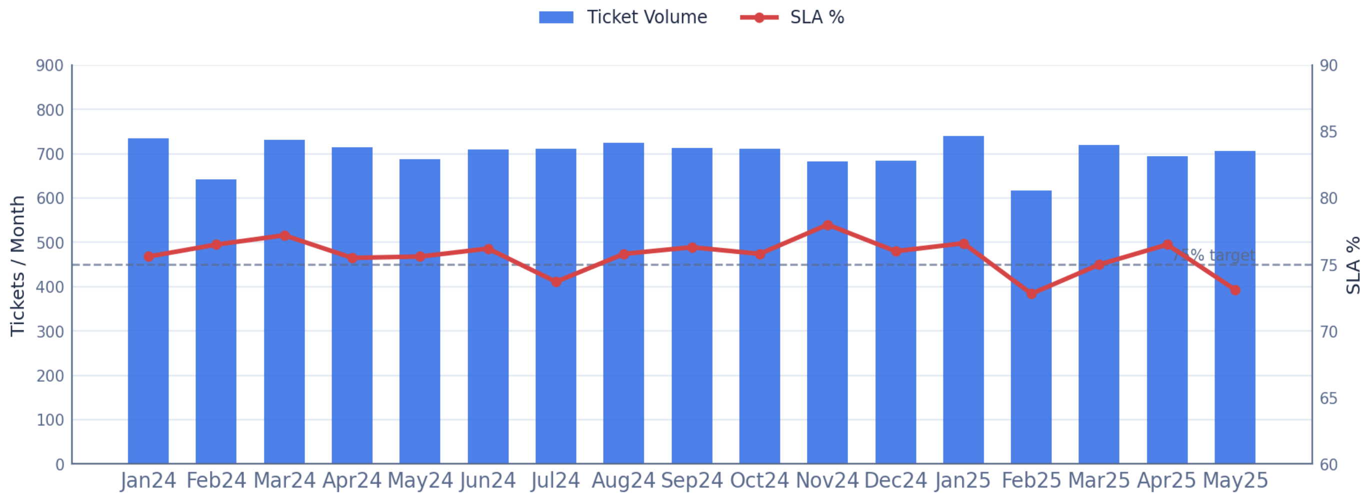

Fig. 1 - Monthly ticket volume (bars) vs. SLA % (line), Jan 2024 - May 2025

> SLA was achieved in 14 of the 17 months, missing target in **July 2024, February 2025, and May 2025** — each worth a quick root-cause check against staffing schedules for that period.

5 Priority Analysis

Tickets are split across three priority levels, each with its own SLA target. Volume and SLA performance move in opposite directions here — the busiest priority tier is also the one missing target by the widest margin.

Fig. 2 - Ticket volume and SLA compliance by priority

Priority	Tickets	Within SLA	SLA %	Violations	Status
High	4,571	4,494	98.3%	77	Meeting Target
Medium	4,952	2,775	56.0%	2,177	Below Target
Low	2,400	1,754	73.1%	646	Close to Target
Overall	**11,923**	**9,023**	**75.7%**	**2,900**	**Meeting Target**

Medium-priority tickets are 42% of total volume but only 56.0% SLA-compliant — they account for roughly **three-quarters of all 2,900 SLA violations**, despite having a longer 72-hour target than High priority. This is the single biggest lever for improving overall SLA performance.

6 Ticket Type Analysis

Tickets are also classified by type — Incident, Request, Problem, and Change — reflecting whether work is reactive (fixing something broken) or proactive (planned changes, service requests).

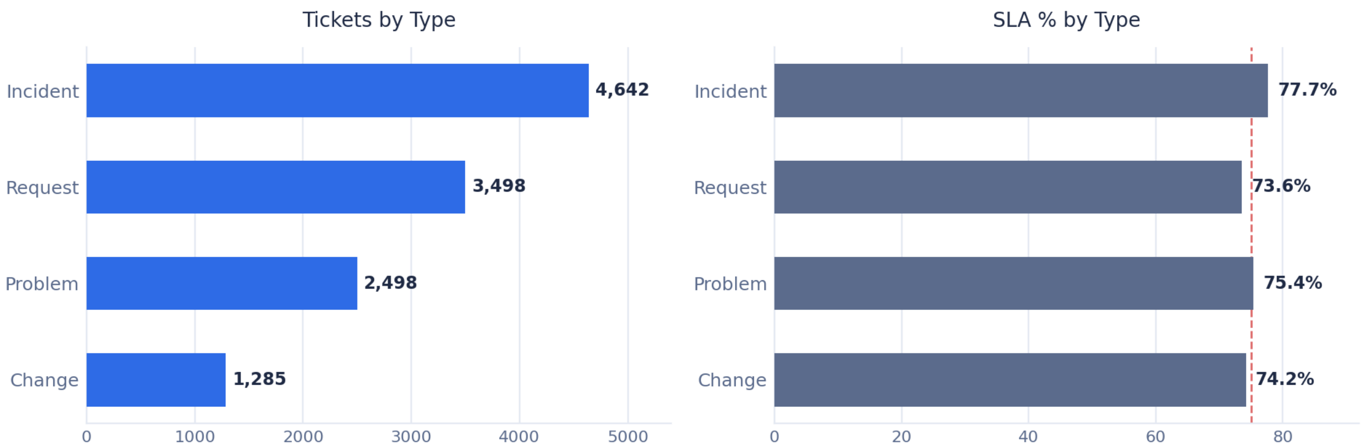

Fig. 3 - Ticket volume and SLA % by type

Incidents (4,642) and Requests (3,498) together make up 68% of all tickets, versus Problems (2,498) and Changes (1,285) — the team spends most of its time on reactive fixes and day-to-day requests rather than root-cause Problem work or planned Change activity. SLA performance by type is fairly tight (73.6%-77.7%), a much narrower spread than the priority-level gap — confirming priority, not ticket type, is the dominant factor behind SLA misses.

7 Queue & Workload Analysis

Tickets route into 10 queues by function. Volume is heavily concentrated in a handful of queues, which is useful for prioritizing staffing and process-improvement investment.

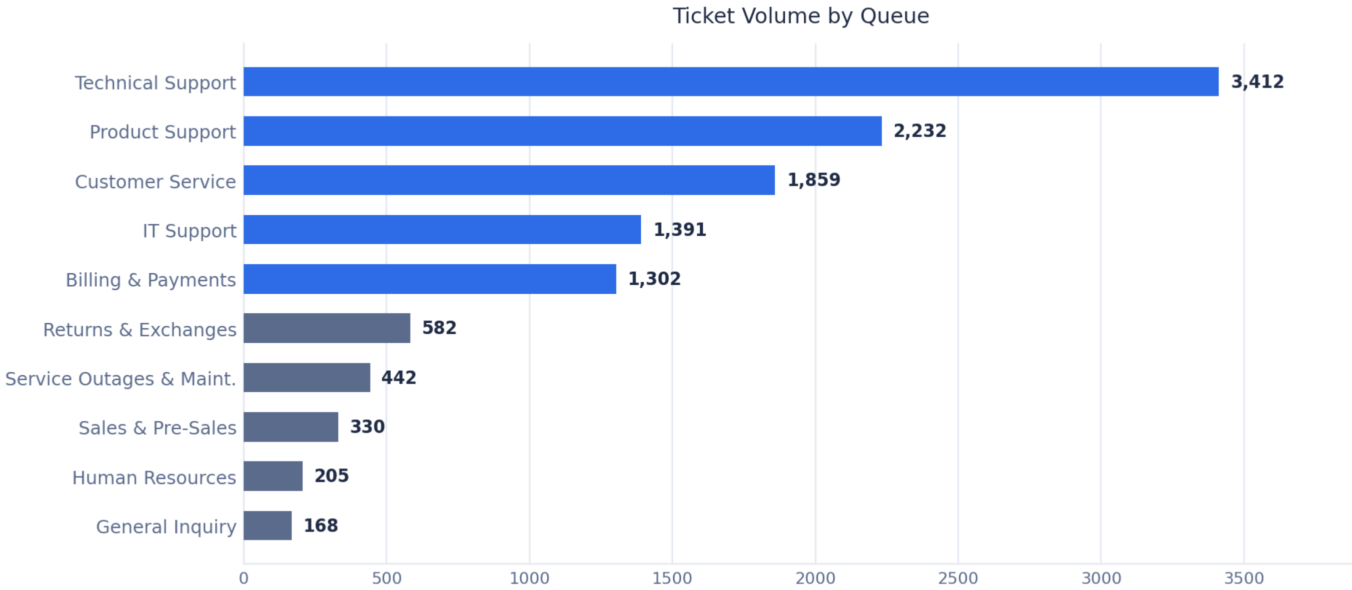

Fig. 4 - Ticket volume by queue

Technical Support (3,412) and Product Support (2,232) together account for roughly **half of all ticket volume** across the 10 queues — the highest-leverage queues for additional staffing or process improvement. Service Outages & Maintenance has the smallest volume (442) but the highest SLA compliance (90.3%), likely reflecting a smaller, more specialized team.

8 Geographic Distribution

Ticket volume is spread evenly across the 10 countries served (1,137-1,240 tickets each), so no single market dominates demand. SLA performance is the more interesting cut here.

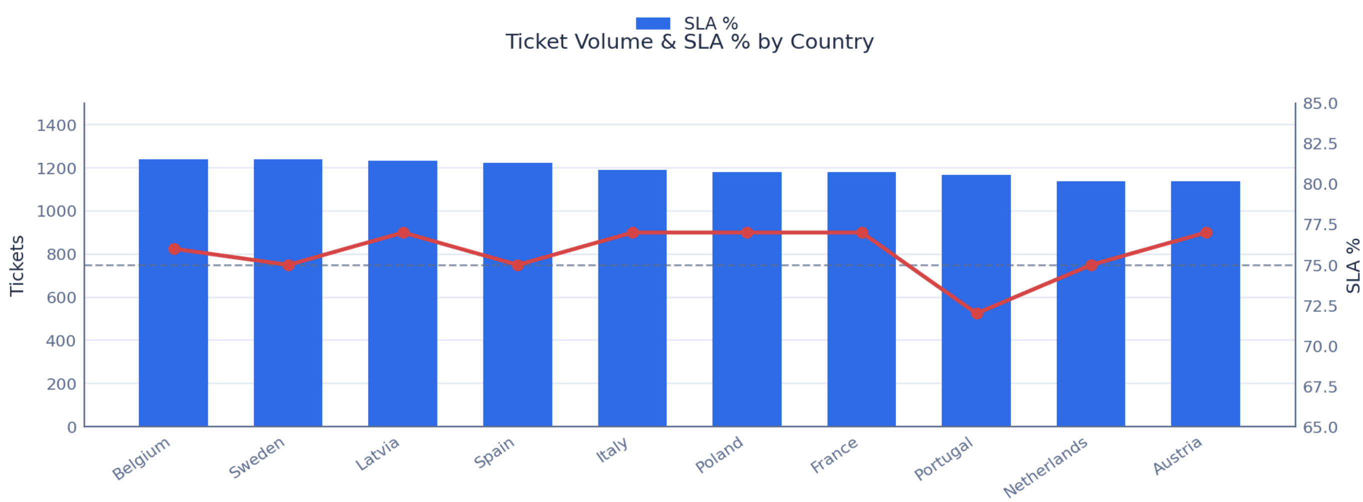

Fig. 5 - Ticket volume and SLA % by country

7 of 10 countries met the 75% SLA target. Portugal is the clearest outlier at 72%; Sweden and the Netherlands sit right at 75%. Overall, country-level SLA compliance is tightly clustered (72%-77%) — a narrow enough spread to conclude

that geography is not the primary driver of missed SLAs. That distinction belongs to priority level (Section 5) and, as the next section shows, to the individual agent.

9 Agent Performance

12 agents handle the full ticket load, from 156 to 2,853 tickets each. Plotting SLA adherence against customer rating side by side shows the two metrics do not move together — some agents are fast but rated lower, others are highly rated but slower to close tickets within SLA.

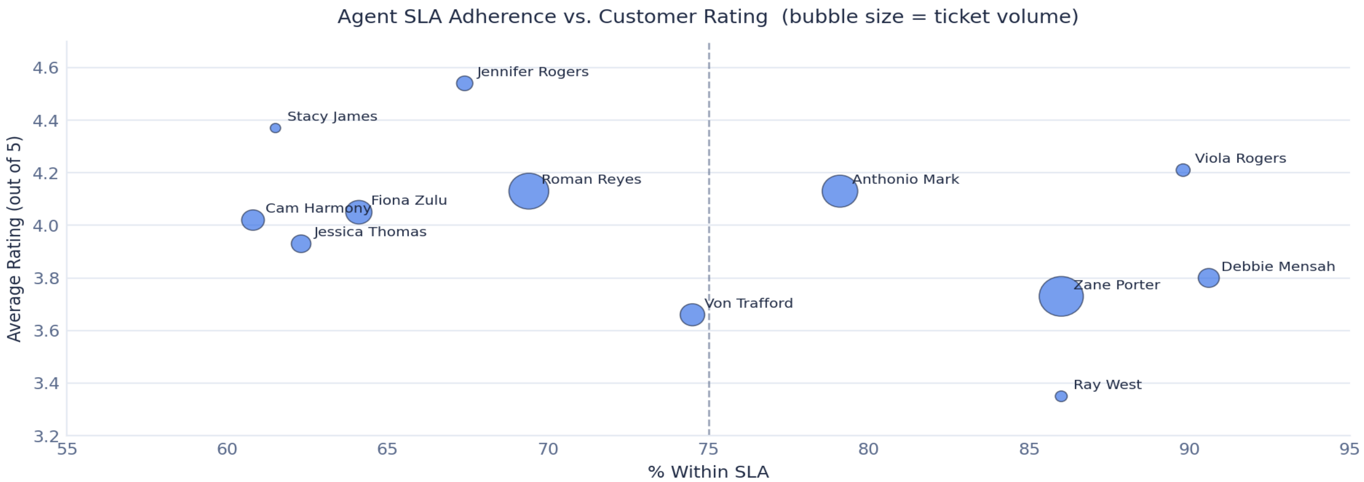

Fig. 6 - Agent SLA adherence vs. average rating (bubble size = ticket volume)

Agent	Tickets	Avg Rating	% Within SLA
Zane Porter	2,853	3.73	86.0%
Debbie Mensah	647	3.80	90.6%
Viola Rogers	284	4.21	89.8%
Anthonio Mark	1,857	4.13	79.1%
Von Trafford	881	3.66	74.5%
Roman Reyes	2,325	4.13	69.4%
Jennifer Rogers	389	4.54	67.4%
Fiona Zulu	1,007	4.05	64.1%
Cam Harmony	757	4.02	60.8%
Jessica Thomas	560	3.93	62.3%
Stacy James	156	4.37	61.5%
Ray West	207	3.35	86.0%

SLA compliance across agents ranges from roughly **61% to 91% — a ~30-point spread**, wider than the gap seen across countries or ticket types, and among agents handling broadly similar ticket mixes. This makes agent-level coaching and workload balancing the highest-leverage people lever in the data.

10 Customer Satisfaction

All 11,923 tickets carry a customer rating, averaging 3.97 out of 5. Looking at the full distribution (rather than just the average) shows the shape of customer sentiment.

Fig. 7 - Distribution of customer ratings across all tickets

Ratings skew positive: 2.5-star ratings are the smallest group (911 tickets, 7.6%), while 5-star ratings are the largest single group (2,837 tickets, 23.8%). Combined, 4-star-and-above ratings make up just under 60% of all feedback. As shown in Section 9, satisfaction does not track SLA adherence in lockstep — both are worth tracking as separate agent-coaching metrics rather than one combined "performance score."

11 Key Findings & Recommendations

11.1 Key Findings

1. Medium priority is the SLA problem, not the average.

Medium-priority tickets are 42% of all volume and only 56.0% SLA-compliant, contributing roughly 75% of all SLA violations despite a longer (72-hour) target than High priority.

2. Three specific months broke an otherwise stable trend.

SLA was met in 14 of 17 months; July 2024, February 2025, and May 2025 missed target and are good candidates for a root-cause review.

3. Geography is not the driver of missed SLAs — agents are.

Country-level SLA compliance is tightly clustered (72%-77%), while agent-level compliance spans roughly 61%-91%, a much wider gap pointing to individual coaching opportunities.

4. Ticket type has a smaller, secondary effect.

SLA-by-type ranges narrowly from 73.6% to 77.7% — real, but far smaller than the priority-level split, confirming priority is the dominant lever.

5. Speed and satisfaction are distinct metrics.

High-rating agents are not always the fastest, and vice versa — both should be coached separately rather than assumed to move together.

6. Two queues carry roughly half the workload.

Technical Support and Product Support account for close to half of all ticket volume across 10 queues, making them the highest-leverage queues for staffing decisions.

11.2 Recommendations

1. Prioritize the Medium-priority SLA gap.

Review triage rules and staffing for Medium-priority tickets first — closing even part of the 56% to 75% gap moves overall SLA performance more than any other single change.

2. Root-cause the three off-target months.

Check staffing schedules and ticket-volume data for July 2024, February 2025, and May 2025 to confirm whether the misses were capacity-driven or process-driven.

3. Use the agent SLA spread as a coaching tool.

Pair high-SLA agents (Debbie Mensah 90.6%, Viola Rogers 89.8%, Ray West 86.0%, Zane Porter 86.0%) with lower-adherence peers to share queue-handling practices.

4. Track SLA and satisfaction as separate KPIs in reviews.

The data shows they don't consistently move together, so a single combined score would hide real, actionable differences between speed and service quality.

5. Right-size staffing for the top two queues.

Technical Support and Product Support drive close to half of total ticket volume; confirm these queues are resourced in line with that load before addressing smaller queues.

12 Appendix - Data Dictionary

12.1 Data Dictionary (key columns)

Column	Description
Ticket ID	Unique identifier for each support ticket
Date / Resolution Date	Ticket creation and resolution timestamps; basis for Resolution Hours
Type	Request, Problem, Incident, or Change
Priority	High, Medium, or Low
Queue	Support department/queue the ticket was routed to

Column	Description
Agent	Support agent who handled the ticket
Rating	Customer satisfaction rating (blank if unrated)
Country	Country the ticket originated from
Severity (derived)	Critical / Major / Normal / Minor, from Priority + Type
SLA Status (derived)	Whether Resolution Hours met the Priority SLA target

12.2 Tools & Techniques

Power BI Desktop (data modeling & report build), Power Query / M (data cleaning), DAX (42 measures), star-schema design (fact + Calendar + Country dimensions), bookmark-driven report navigation with dynamic titles.

12.3 Repository

Full project files, the .pbix report, and this report are available in the project's GitHub repository — see README.md for setup instructions.

Prepared by Sreekar Tirunagari — Associate, IT Helpdesk transitioning into Business Analytics.